ADAPTIVE INPUT SHAPING CONTROL OF A ROTARY HOOK

Ho Duc Tho, Kento Takemori, Ryosuke Tasaki, Takanori Miyoshi and Kazuhiko Terashima

Mechanical Engineering Department, Toyohashi University of Technology, Toyohashi, Aichi, Japan
Emails: {ho_duc_tho, takemori}@syscon.me.tut.ac.jp; {tasaki, miyoshi, terasima}@me.tut.ac.jp

Abstract: In crane systems, rotary hook plays an important role in adjusting the skew angle of payload to match with skew angle of vessels or container trucks. At the present, rotary hook is mainly controlled manually by experienced and skilled workers. Therefore, this paper presents to develop an adaptive input shaping (AIS) algorithm for rotary hook system which can automatically drive the payload to desired angle while suppressing the residual oscillation caused by flexible ropes. An algebraic estimator will be the center of adaptive input shaping scheme. Moreover, the comparison between AIS and built-in robust Specified Insensitivity (SI) input shaper will be also conducted when both of them are shaped to be robust with the natural frequency uncertainty of rotary hook system. All of conclusions will be verified through simulation results.

Keywords: Adaptive input shaping, algebraic estimator, rotary hook, boom crane.

1. INTRODUCTION

Orientation control of payload plays an important role in real crane systems. For instance, in boom crane system working at harbor as shown in Fig. 1, payloads now are containers and they could lie on the ground with arbitrary skew angle. Only when the skew angles of containers and vessel are the same, do container loading process is available. It is difficult to make the orientation of the vessel match with skew angle of container on the shore so that the inverse process is reasonable. Similar problem can be encountered in unloading process (Fig. 2) in which the container even must be turned 180^0 in order to make container door at the rare end of the truck. In reality, an equipment called rotary hook will control and correct the skew angle of container to match with the skew angle of vessel or truck. In fact, as shown in Fig. 3 container will be gripped by twist-lock mechanism of spreader, spreader is attached to a motor shaft which is mounted inside the housing of rotary hook and finally the rotary hook will be hung below boom tip by ropes.

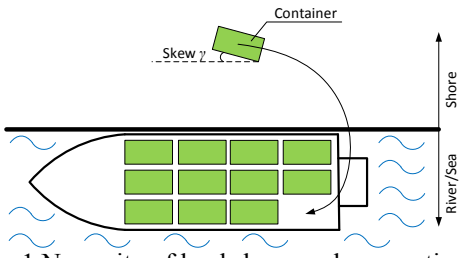

Fig. 1 Necessity of load skew angle correction in loading case.

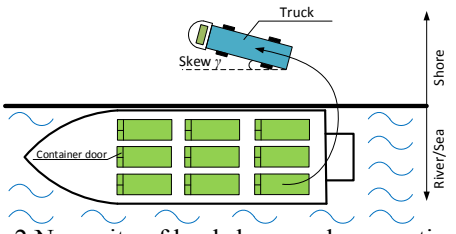

Fig. 2 Necessity of load skew angle correction in unloading case.

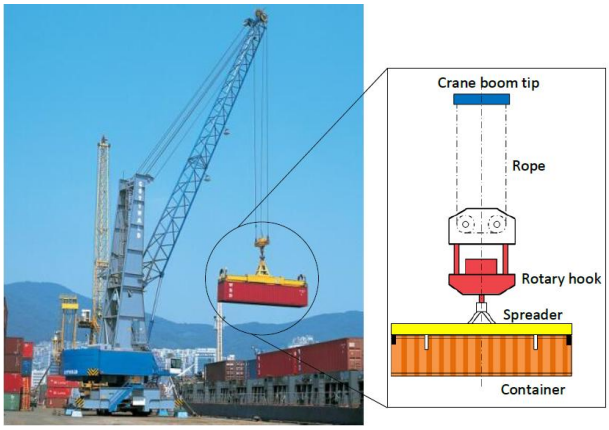

Fig 3. Boom crane working at harbor with rotary hook

Nowadays, payload's skew angle control through rotary hook is always done manually by worker. Because rotary hook is an under-actuated system which has only one control input (i.e. torque τ generated by a motor) and two control outputs (i.e. rotary hook skew angle θ and payload angle φ) hence it requires much experience from the worker and it is inefficient as well. Therefore, developing automatic controllers for rotary hook which can drive the payload to specified angle precisely while suppressing the oscillation of rotary hook skew angle is a natural need of any crane system.

Automatic skew control for rotary hook was first proposed by Gustafsson et al., [1]. This research mainly developed a time optimal controller as well as presented an extended Kalman filter to estimate the absolute angle of hook skew angle based on the measurements of gyroscope and position encoder sensors. Recently, a two-degrees of freedom control concept (2-DOF) is derived which comprises a state observer for the skew dynamics, a reference trajectory generator, and a feedback control law (Schaper et al., [2]). The performance of the control system is validated with test drives on a full-size mobile harbor boom crane. However, none of above researches consider the uncertainty of rotary hook system, especially in payload mass and payload inertia which is the main issue

encountered in reality. Wide variation range of rotary hook system will affect seriously to performance of controllers and even causes instability.

Another simple and effective control strategy for light-damped dynamic system like rotary hook is the input shaping controller. Since the first time input shaping controller with Zero-Vibration (ZV) input shaper and its robust versions (Singer et al., [3-4], Singhose et al., [5]) were proposed, various researches tried to implement this type of controller on overhead crane (Singhose et al., [6], Ngo and Hong et al., [7]) or flexible manipulator (Tzes et al., [8], Park et al., [9]). However, all of aforementioned input shapers have the same characteristic: the more the shapers robust to modelling errors, the slower maneuver is and this will seriously affect to productivity of overall system. Above stumbling block will be overcome by using adaptive input shaping algorithm which enable us to use the fastest input shaper (ZV) while can adapt the uncertainty of system parameters to maintain the residual oscillation eliminating ability.

This paper aims to develop an adaptive input shaping (AIS) controller for rotary hook system in order to drive the payload to desire angle and suppress the residual vibration simultaneously considering the uncertainty in system natural frequency. The main issue in AIS algorithm is the natural frequency online identification process to determine remain unknown parameters of impulses in input shaper. In fact, the performance of AIS is dominated by the choice of identification technique. An AIS can be developed in frequency domain with indirect adaption algorithm where Fast Fourier Transform (FFT) was employed (Tzes et al., [10]). However, this method required high numeric computation and slow parameters convergence. Bodson et al., [11] proposed a direct adaption law for both natural frequency and damping ratio estimation based on a parameterization poles/zeros cancellation controller. Nonetheless, the explicit persistency of reference signal for parameters convergence is the major problem of this technique where constant reference signal is preferred. Recently, Trapero et al., [12] proposed an algebraic estimator to estimate the frequency of biased and noisy sinusoidal signals in signal processing area. This method was successfully applied to vibration monitoring (Trapero et al., [13]) and adaptive input shaping for flexible manipulator (Pereira et al., [14-15]). The simplicity yet effectiveness of this algebraic estimator in identifying the natural frequency of dynamic system has motivated this paper to use it as a part of AIS algorithm for rotary hook system.

The contributions of this paper are as following: first, the dynamic of rotary hook system is further adjusted by accounting for the torque of motor acting on the payload. Second, AIS controller utilizing algebraic estimator is proposed for under-actuated rotary hook system. Third, the comparison between AIS and Specified Insensitivity - a built-in robust input shaper is carried out and it is later recognized that AIS shows superior performance. All of conclusions are verified by simulation results.

2. ROTARY HOOK MODELLING

This section provides brief description on the model of a rotary hook system. The modelling of rotary hook in this paper is a modification of the work found in Gustafsson et al., [1] by considering the involvement of motor torque acting on the load. Fig. 4 showed the skew motion of the rotary hook system resulted from the rotation movement of payload. The definition of parameters used in the rotary hook system is illustrated in Table 1.

Using Euler-Lagrange method, the linear dynamic of rotary hook system is described by Eq. (1) – (2):

$$I_L\ddot{\varphi} = \tau \tag{1}$$

$$\ddot{\theta} = \frac{-mR^2 g}{l\left(I_L + I_H\right)}\theta - \frac{c}{I_L + I_H}\dot{\theta} + \frac{1}{I_L + I_H}\tau \tag{2}$$

The natural frequency of the rotary hook system is given as following:

$$\omega_n = \sqrt{\frac{mR^2 g}{l\left(I_L + I_H\right)}} \tag{3}$$

It is should be noted that Eq. (1) and (2) which describe the dynamic of rotary hook system will be used for algebraic estimator to online identify the natural frequency given by Eq. (3). In reality, only rotary hook inertia I_H and viscous damping c are assumed to be known. Besides, it is observed that rotary hook is an under-actuated system which has only one control input $\tau(t)$ and two control outputs ($\varphi(t)$, $\theta(t)$).

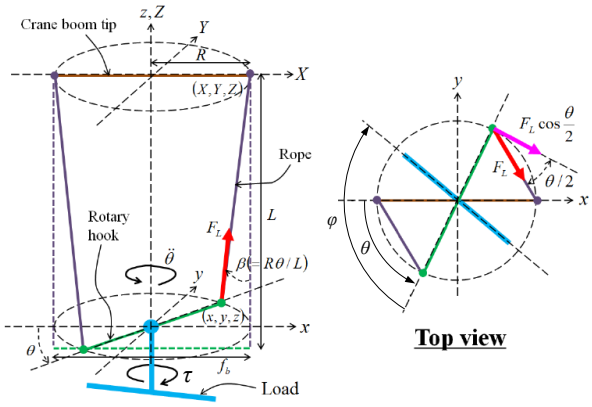

Fig. 4 Rotary hook modelling

Table 1. Parameters of rotary hook system

Parameter	Description
$\varphi(t)$	Relative angle between the payload and the hook (rad)
φ_d	Reference angle of $\varphi(t)$
$\theta(t)$	Absolute angle of rotary hook versus the x axis (rad)
$\tau(t)$	Torque generated by motor (Nm)
I_L	Inertia of load (kgm^2)
I_H	Inertia of rotary hook (kgm^2)
m	Total mass of load and hook (kg)
R	Skewing radius (m)
l	Rope length (m)
c	Friction coefficient (Ns/rad)
g	Gravity acceleration (m/s^2)

In real implementation, the relative angle between the load and the hook ($\varphi(t)$) can be easily obtained by reading the encoder attached at the end of servo motor, whereas an Inertia Measurement Unit (IMU) must be utilized to obtain the absolute angle of the hook versus x axis ($\theta(t)$). Furthermore, in reality, the crane operator presses the left button and right button to command the desired payload angle φ_d, the motion controller will receive this reference load angle and modulated it by adaptive input shaping technique to formulate the shaped command. Shaped command becomes the actual reference trajectory for payload angle. The following sessions are discussed on AIS algorithm.

3. ADAPTIVE INPUT SHAPING SCHEME

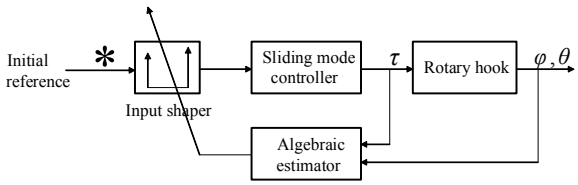

Fig. 5 Adaptive input shaping algorithm

Consider the impulses sequence of standard non-robust ZV input shaper for the system which has natural frequency ω_n and damping ratio ξ, the time location (t_i) and magnitude (A_i) of each impulse is given as following matrix:

$$\begin{bmatrix} A_j \\ t_j \end{bmatrix} = \begin{bmatrix} \dfrac{1}{1+K} & \dfrac{K}{1+K} \\ 0 & 0.5T_d \end{bmatrix}, \tag{4}$$

where $K = \exp\left(\dfrac{-\xi\pi}{\sqrt{1-\xi^2}}\right)$ and $T_d = \dfrac{\pi}{\omega_n\sqrt{1-\xi^2}}$. For light-damped system, we have $\xi \approx 0$ hence $K = 1$, $T_d = \pi/\omega_n$. Therefore, the only unknown in impulses parameters of input shaper is the natural frequency ω_n and this motivates the development of adaptive algorithm which can online estimate this value by an identification technique. For the sake of simplicity and fast parameter convergence, an algebraic estimator will be implemented.

The AIS algorithm shown in Fig. 5 can be illustrated as following: initial reference payload angle (φ_d) will be convolved with the known first impulse of input shaper then the payload is controlled to track the 1st shaped command by a robust sliding mode controller. During the motion of payload, the values of control input (τ) and control outputs (φ and θ) are employed by algebraic estimator to online estimate the natural frequency and when the estimation is converged, estimated value of natural frequency will be used to update the complete payload reference trajectory. Denote the convergence value of natural frequency is $\omega_{converge}$ and convergence time is t_e, it is preferable when t_e is less than the time location of 2nd impulse which is determined by $\pi/\omega_{converge}$ (let call this value is *critical time*). In the worse cases, if t_e is larger than critical time then the amount $\Delta t = t_e - \pi/\omega_{converge}$ can be considered as modelling error and it can be easily solved by standard robust input shaper.

4. ALGEBRAIC ESTIMATOR

4.1 Mikusinski operational calculus

In general, algebraic estimator is based on Mikusinski operational calculus in order to solve the ordinary differential equations to extract the necessary parameters. Similar to well-known Laplace transformation, by using Mikusinski operator the differential equation can be transformed into algebraic equation in which it can be solved easily.

Let $\mathfrak{C}$ denotes a ring of all real and complex – valued continuous functions of real variable t, defined in the interval $[0, \infty)$. For two functions $f, g \in \mathfrak{C}$, define two binary operators: addition and multiplication as following [16]:

$$f + g = \{f(t) + g(t)\} \tag{5}$$

$$fg = \left\{\int_0^t (f-u)g(u)du\right\} \tag{6}$$

where f or $\{f(t)\}$ is a function and $f(t)$ means the value at time t of function f.

Assign h as a constant function $\{1\}$, this is also so-called *integral operator*, for any $f \in \mathfrak{C}$ we have:

$$hf = \left\{\int_0^t f(u)du\right\} \tag{7}$$

A fundamental role in Mikusinski operational calculus is played by differential operator which is defined as $s = 1/h$. If a function $f = \{f(t)\}$ has n^{th} derivative $f^{(n)} = \{f^{(n)}(t)\}$ continuous in the interval $[0, \infty)$ then following equations hold:

$$f^{(n)} = s^n f - s^{n-1} f(0) - s^{n-2} f^{(1)}(0) - \ldots - s f^{(n-2)}(0) - f^{(n-1)}(0) \tag{8}$$

$$\frac{d^k}{ds^k} f = (-1)^k t^k f \tag{9}$$

Eq. (8) and (9) will be used to transform the differential equations in Eq. (1) and (2) which describe the dynamic of rotary hook system to derive necessary parameters.

4.2 Load inertia estimation

First, Mikusinski operator will be implemented to estimate the load inertia (I_L) by using the first rotary hook dynamic equation:

$$\begin{aligned} & I_L\ddot{\varphi} = \tau \\ & \Rightarrow I_L\left[s^2\varphi(t) - s\varphi(0) - \dot{\varphi}(0)\right] = \tau \end{aligned} \tag{10}$$

In order to eliminate the effect of initial condition on estimation, differentiate Eq. (10) two times, we have:

$$I_L\left[2\varphi(t) + 4s\frac{d\varphi(t)}{ds} + s^2\frac{d^2\varphi(t)}{ds^2}\right] = \frac{d^2\tau}{ds^2} \tag{11}$$

Multiplying both sides of Eq. (11) with s^{-2} in order to make *integral operator* appear, the result is as following:

$$I_L\left[2s^{-2}\varphi(t) + 4s^{-1}\frac{d\varphi(t)}{ds} + \frac{d^2\varphi(t)}{ds^2}\right] = s^{-2}\frac{d^2\tau}{ds^2} \tag{12}$$

Therefore, load inertia can be derived as:

$$I_L = \frac{s^{-2}\dfrac{d^2\tau}{ds^2}}{2s^{-2}\varphi(t) + 4s^{-1}\dfrac{d\varphi(t)}{ds} + \dfrac{d^2\varphi(t)}{ds^2}} \tag{13}$$

Apply the result in Eq. (9), we have:

$$\frac{d\varphi(t)}{ds} = -t\varphi(t); \frac{d^2\varphi(t)}{ds^2} = t^2\varphi(t); \frac{d^2\tau(t)}{ds^2} = t^2\tau(t) \tag{14}$$

Hence, the load inertia can be further written as:

$$I_L = \frac{\int_0^t\int_0^\sigma \lambda^2\tau(\lambda)d\lambda d\sigma}{2\int_0^t\int_0^\sigma \varphi(\lambda)d\lambda d\sigma - 4\int_0^t \sigma\varphi(\sigma)d\sigma + t^2\varphi(t)} \tag{15}$$

In practice, Eq. (15) cannot be used for real time calculation so that it is reasonable to transform it into a set of differential equations for online computation. Assign:

$$\begin{aligned} &n_1(t) = x_1;\ \dot{x}_1 = x_2;\ \dot{x}_2 = t^2\tau(t) \\ &d_1(t) = t^2\varphi(t) + x_3;\ \dot{x}_3 = x_4 - 4t\varphi(t);\ \dot{x}_4 = 2\varphi(t) \end{aligned} \tag{16}$$

Furthermore, in order to deal with noise, a second order low-pass filter which has cut-off frequency ω_c and damping ratio ε will be applied on both numerator $n_1(t)$ and denominator $d_1(t)$ [13] and define new variables as following:

$$\begin{aligned} &n_{1f}(t) = x_5;\ \dot{x}_5 = x_6;\ \dot{x}_6 = -2\varepsilon\omega_c x_6 - \omega_c^2(x_5 - x_1) \\ &d_{1f}(t) = x_7;\ \dot{x}_7 = x_8;\ \dot{x}_8 = -2\varepsilon\omega_c x_8 - \omega_c^2(x_7 - x_3 - t^2\varphi(t)) \end{aligned} \tag{17}$$

The load inertia estimation value I_{Le} can be finally given as below:

$$I_{Le} = \frac{n_{1f}(t)}{d_{1f}(t)} = \begin{cases} \text{arbitrary}, t \in (0, \delta_1) \\ \dfrac{x_5}{x_7},\ t \in [\delta_1, \infty) \end{cases} \tag{18}$$

4.3 Natural frequency estimation

After obtaining the estimation value of load inertia, the only unknown parameter in Eq. (2) is the natural frequency hence Mikusinski operator now can be also applied on this equation. By implementing the similar procedure with load inertia estimation, a set of differential equations corresponding to the solution of estimated natural frequency is given in Eq. (19):

$$\begin{cases} \dot{x}_1 = \dfrac{1}{I_L + I_H}x_2 - 2x_3 + 4t\theta(t) \\ \qquad - \dfrac{c}{I_L + I_H}\left[-2x_4 + t^2\theta(t)\right] \\ \dot{x}_2 = t^2\tau(t);\ \dot{x}_3 = \theta(t);\ \dot{x}_4 = t\theta(t) \\ \dot{x}_5 = x_6;\ \dot{x}_6 = t^2\theta(t) \\ \dot{x}_7 = x_8; \\ \dot{x}_8 = -2\varepsilon\omega_c x_8 - \omega_c^2\left[x_7 - x_1 + t^2\theta(t)\right] \\ \dot{x}_9 = x_{10} \\ \dot{x}_{10} = -2\varepsilon\omega_c x_{10} - \omega_c^2(x_9 - x_5) \end{cases} \tag{19}$$

The estimation value of natural frequency ω_e now can be derived as:

$$\omega_e = \begin{cases} \text{arbitrary}, t \in (0, \delta_1 + \delta_2) \\ \sqrt{\dfrac{x_7}{x_9}},\ t \in [\delta_1 + \delta_2, \infty) \end{cases} \tag{20}$$

It should be noted that at a very first small time interval δ_1, the estimation value of load inertia in Eq. (18) is ill-defined, however beyond this interval the estimated load inertia value will converge to its real value. In addition, the estimation value of natural frequency only start to converge after a small interval $\delta_1 + \delta_2$ because the load inertia and natural frequency estimations are serial processes. In practice, sets of equations (16) + (17) of load inertia estimation and (19) of natural frequency estimation can be online computed by many types of fixed-step ODE solvers. In particular, the Runge-Kutta is a solver which can compromise between the computational burden and accuracy.

In order to demonstrate the effectiveness of algebraic estimator, simulation study will be conducted under following reference parameters: I_L = 16.491 kgm^2, I_H = 1.127 kgm^2, m = 142.19 kg, R = 0.25 m, l = 4 m, c = 0.1 Ns/rad, g = 9.81 m/s^2. For smooth motion, the chosen initial reference profile of payload angle is a 2nd order trajectory as shown in Eq. (21):

$$r(t) = \begin{cases} \dfrac{4x_f}{t_f^2}\dfrac{t^2}{2}\ \ (0 \le t \le \dfrac{t_f}{2}) \\ \dfrac{x_f}{t_f^2}(-2t^2 + 4t_f t - t_f^2)\ \ (t \ge \dfrac{t_f}{2}) \end{cases} \tag{21}$$

A sliding mode controller is then designed to control the payload tracking the initial reference trajectory (where chosen of x_f and t_f are 90^0 and 5 second respectively). Three measured signals including load angle (φ), rotary hook skew angle (θ) and torque acting on the load (τ) will be feed into algebraic estimator to estimate the natural frequency of system. The result of payload inertia and natural frequency estimation is illustrated in Fig. 6.

4.3 Algebraic estimator convergence criterion

Moving average convergence criterion is chosen to determine when the estimation converges. This type of criterion is demonstrated as following [13]:

$$\frac{MA(k) - MA(k-M)}{MA(k)} < tol$$

$$\text{where, } MA(k) = \frac{\sum_{i=0}^{M}\omega_e(k-i)}{M+1},\ k > 2M+1 \tag{22}$$

M : window number

In fact, this criterion will seek for the time when the estimation curve is as flat as desire level which is determined by both window number M and tolerance tol. It should be noted that when the window number is large and tolerance is small then the estimation error will be small but convergence time is large and vice versa. Therefore, some trial and errors should be performed to determine the best M and tol value that

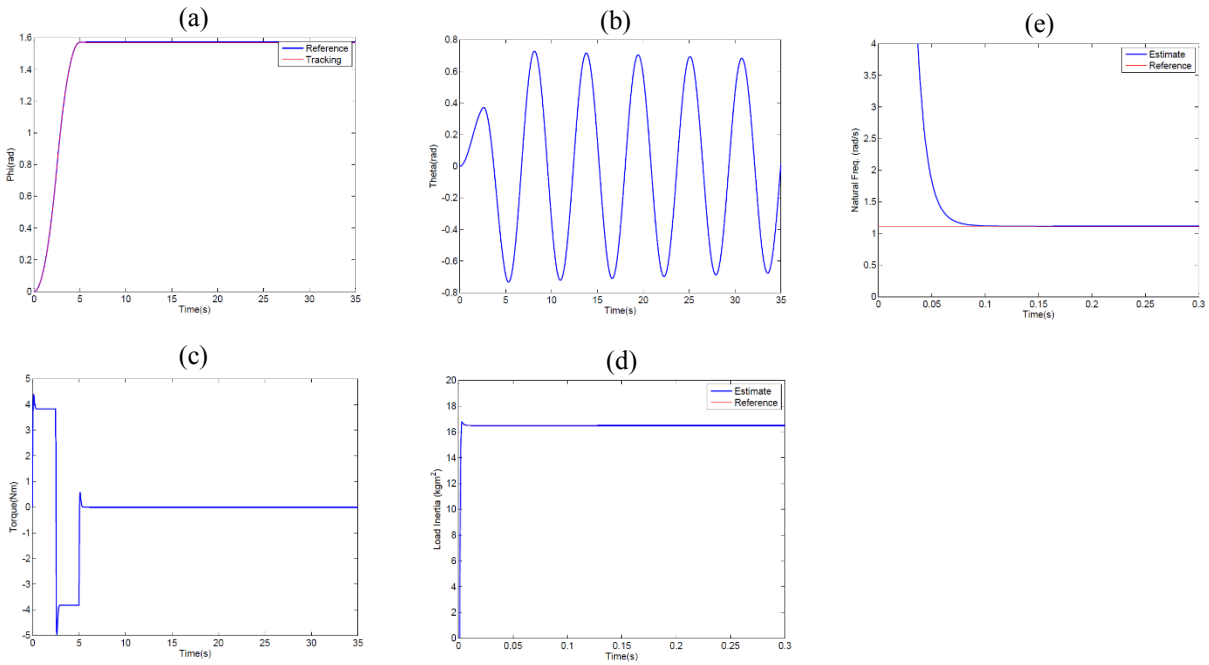

Fig. 6 Algebraic estimator simulation result: (a) Payload angle response, (b) Rotary hook skew angle response, (c) Torque acting on the load, (d) Load inertia estimation result, (e) Natural frequency estimation result.

compromise between convergence time and estimation error.

Moreover, the precision of algebraic estimator is measured by Mean Absolute Error Percentage (MAEP) which is given by:

$$\text{MAEP} = \frac{\left|\omega_a - \omega_{converge}\right|}{\omega_a}.100\% \tag{23}$$

where ω_a is the actual value and $\omega_{converge}$ is the converged value of system natural frequency.

The chosen value of M and tol are 150 and 1e-7 will be used throughout the rest of paper. By using these values, the convergence time and MAEP of algebraic estimator in previous simulation study is only 0.2428 second and $3.85.10^{-4}$% respectively.

5. MONTE CARLO SIMULATION

As it was mentioned earlier, in reality the rotary hook parameters including payload mass, payload inertia and skewing radius vary in wide ranges which may cause the instability to the controllers. In this study, these variation ranges are chosen as following: $I_L \in [2.64, 41.89]$ kgm^2, $m \in [77.39, 207]$ kg, $R \in [0.1, 0.4]$ m. The system nominal values are chosen at the average of each interval. Note that aforementioned hook parameters vary more than $\pm$60% from nominal values. In fact, the adaptive input shaping can adapt to any variation range of hook parameters.

Monte Carlo simulation is usually used when there are random variables involve into the process and for this reason, it reasonably suits for algebraic estimator performance evaluation because in reality, the payload specifications are often random. The procedure of Monte Carlo simulation is described as following: for each sample, a set of values I_L, m, R will be randomly generated within the defined interval, the sliding mode controller is used to control the load tracking the reference trajectory and algebraic estimator is then utilized to estimate the natural frequency of each case. Total number of samples are 270. The result of Monte Carlo simulation in both convergence time and estimation error are shown in Fig.7 and 8 respectively.

In the Fig. 7, the convergence time of all samples are less than critical time which mean the payload reference profile can be updated before the time location of second impulse in input shaper sequence. The maximum MAEP of all cases is only 0.0112% (Fig. 8) therefore the non-robust and fastest input shaper – ZV can be chosen to guarantee the vibration percentage less than 5% according to the sensitivity curves of three types of input shaper: ZV/ZVD/ZVDD shown in Fig. 9.

6. SIMULATION RESULTS – COMPARISON BETWEEN AIS AND SI INPUT SHAPER

Specified Insensitivity (SI) input shaper proposed by Singer et al., [4] is a powerful robust input shaper. This type of input shaper can be designed to be robust with any range of system natural frequency so that it will be used to compare with AIS which utilizes a non-robust input shaper in both execution time and residual vibration after the motion.

SI input shaper which is robust to aforementioned variation range of rotary hook parameters can be implicit derived by General Algebraic Modelling System (GAMS) optimizing software. The result is shown in below matrix and the sensitivity curve of designed SI input shaper is described in Fig. 10:

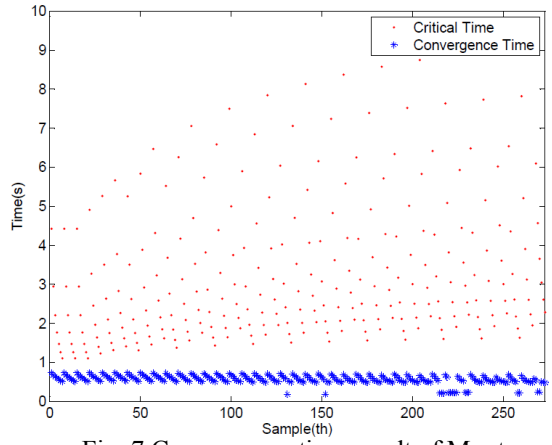

Fig. 7 Convergence time result of Monte Carlo simulation

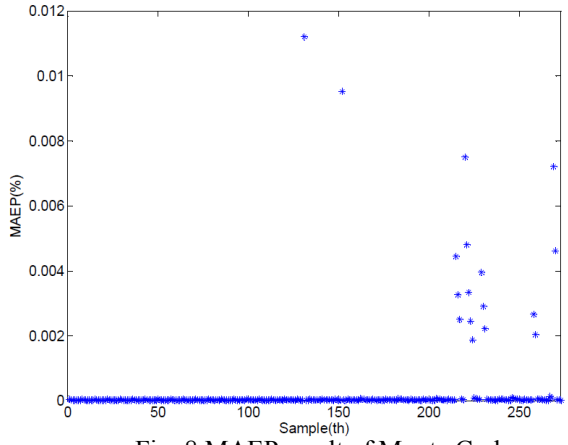

Fig. 8 MAEP result of Monte Carlo simulation

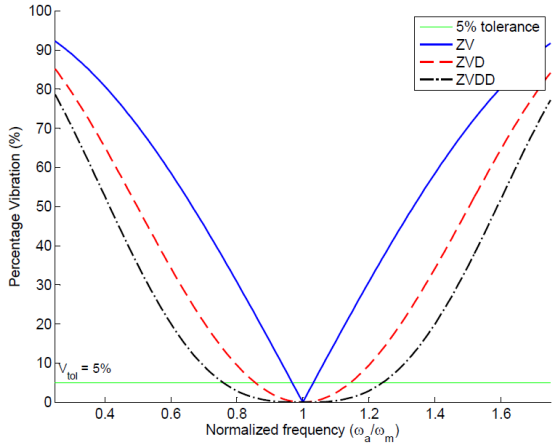

Fig. 9 Sensitivity curves of ZV/ZVD/ZVDD input shapers

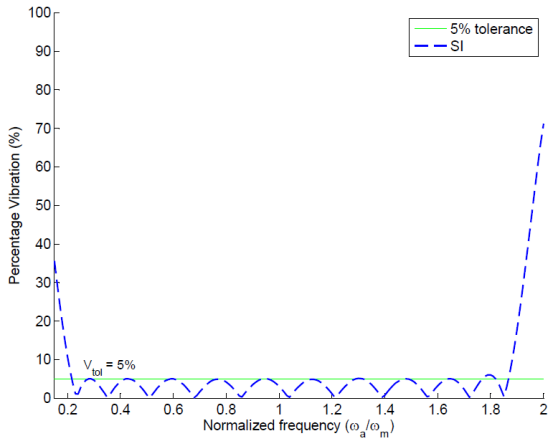

Fig. 10 Sensitivity curves of SI input shapers

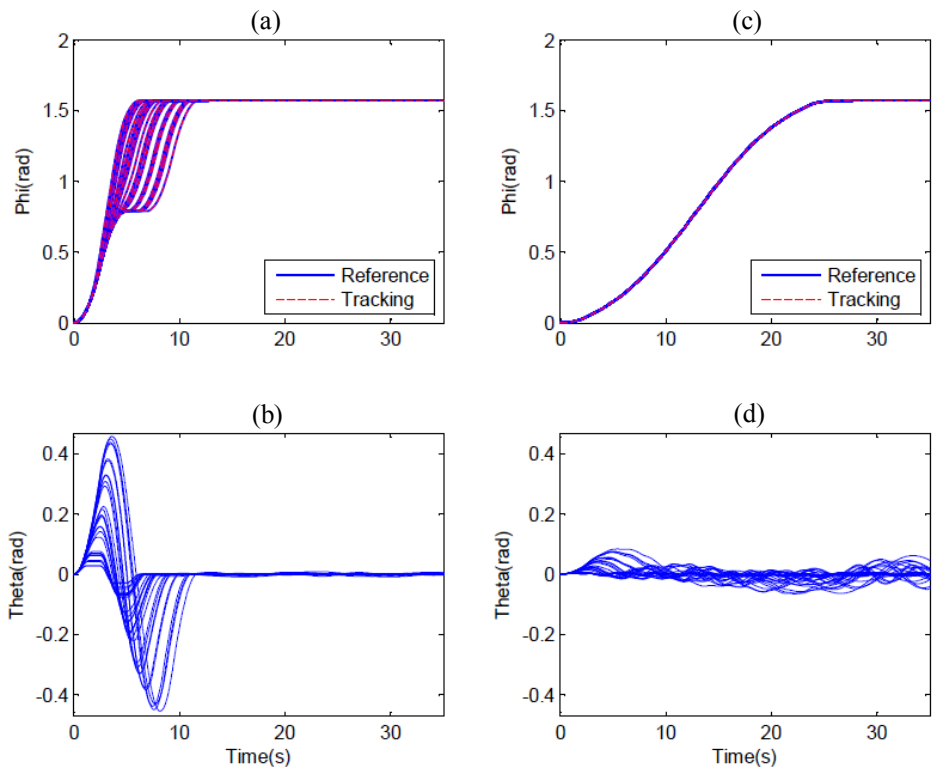

Fig. 11 Simulation results of AIS and SI input shaper: (a) Payload angle response of AIS, (b) Rotary hook skew angle response of AIS, (c) Payload angle response of SI, (d) Rotary hook skew angle response of SI.

$$\begin{bmatrix} t_i \\ A_i \end{bmatrix} = \begin{bmatrix} 0 & 1.932 & 3.836 & 5.737 & 7.637 & 9.532 & 11.43 & 11.43 & 15.22 & 17.12 & 19.02 & 20.93 \\ 0.049 & 0.054 & 0.076 & 0.096 & 0.111 & 0.119 & 0.119 & 0.110 & 0.094 & 0.074 & 0.052 & 0.046 \end{bmatrix} \quad (24)$$

where the first row is time location and second row is amplitude of each impulse in SI input shaper.

Fig. 11 showed the simulation results of AIS and SI input shaper with 30 random cases of hook parameters when the payload desire angle is 90^0. The required time for payload to reach 90^0 in the case of AIS is only about 11.5s in maximum while SI input shaper needs constant 25.93s to finish the motion. Furthermore, the oscillation of rotary hook skew angle (θ) when AIS is employed is much better than SI input shaper which once again indicate the precision of natural frequency estimation using algebraic estimator.

In conclusions, through the simulation results of Fig. 11, AIS showed excellent performance in comparison with standard built-in robust SI input shaper in the viewpoints of transfer time and complete residual vibration suppression ability.

7. CONCLUSIONS

This paper have developed a simple yet effective adaptive input shaping algorithm which showed superior performance in comparison with standard built-in robust SI input shaper when they are both shaped to be robust with a large parameters variation which is often encountered in real rotary hook system. The effectiveness of proposed technique was verified by simulation results. Future work of this research is implementing the aforementioned adaptive input shaping scheme to a miniature rotary hook system which is being constructed at Toyohashi University of Technology.

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